Journal of Environmental Psychology 109 (2026) 102896

Contents lists available at ScienceDirect

Journal of Environmental Psychology

journal homepage: www.sciencedirect.com/journal/journal-of-environmental-psychology

The role of information source in climate beliefs, behavioral commitments, and policy preferences

Danielle Goldwert [a,*], Madalina Vlasceanu [b]

[a] Department of Psychology, New York University, New York, NY, USA
[b] Department of Environmental Social Sciences, Stanford Doerr School of Sustainability, Stanford, CA, USA

ARTICLE INFO

Keywords:
Climate change
Source
Beliefs
Policy
Behavior

ABSTRACT

Source attribution can play a critical role in the credibility, acceptance, and incorporation of information, especially in polarized contexts. Here, we experimentally test how 12 different information sources impact the credibility of climate change information, and result in the incorporation of climate information into the beliefs, behavioral commitments, and policy preferences of people varying along political ideologies. In a sample of 9076 U.S. residents recruited on Prolific, we found that source credibility strongly influences climate beliefs, behavioral commitments, and policy preferences. We also found that scientists and grassroots community advocates are rated as the most credible sources of climate information, and fossil fuel companies or Republican political leaders as the least credible. However, despite differences in source credibility, information provided by different sources was not differentially incorporated into participants' belief systems, nor did it differentially influence their behavioral commitments or policy preferences. We discuss these findings in the context of climate communication efforts.

Stimulating the acquisition and incorporation of accurate information is critical to solving society's hardest collective action problems, ranging from pandemics to climate change. Accordingly, a growing body of work across the behavioral sciences has been investigating the psychological roots of information acceptance and incorporation into climate-relevant beliefs and behaviors – what we refer to as climate information incorporation (Goldwert & Vlasceanu, 2025; Vlasceanu et al., 2023). While most of this literature has focused on the content or framing of the disseminated information (Goldwert, Patel, et al., 2025; Vlasceanu, Doell et al., 2024), less is known about the impact of information sources on the effectiveness of informational interventions in collective action problems. Here, we investigate how different information sources shape climate information incorporation.

Prior work underscored the central role of source credibility in shaping information acceptance (Hovland & Weiss, 1951; Kim & Kim, 2013; Brewer & Ley, 2013; Vlasceanu & Coman, 2022). According to dual-process models of persuasion, such as the Elaboration Likelihood Model (Petty & Cacioppo, 1986), source characteristics can influence persuasion through either the central or peripheral route. Crucially, the Elaboration Likelihood Model does not assume that source cues are inherently peripheral. Rather, the role of source credibility depends on the audience's motivation and ability to process the message, as well as the perceived diagnostic value of the source (Petty, Cacioppo, & Heesacker, 1987). When the source is viewed as a meaningful or relevant argument (e.g., "This expert has knowledge that directly informs the issue"), source credibility can function as a central cue. In other cases, it may serve as a heuristic or peripheral signal. This multiple-roles hypothesis makes it especially important to empirically examine when and how source cues influence belief systems and behavior, particularly in politically polarized contexts like climate change.

Beyond source credibility, other peripheral cues, including perceived expertise, trustworthiness, or social identity alignment, often determine whether information is accepted or rejected (Chaiken, 1980; Hart & Nisbet, 2012; Kahan et al., 2011). For instance, information delivered by in-group or ideologically aligned communicators is reliably viewed as credible and persuasive (Cohen, 2003; Fielding & Hornsey, 2016; Fielding, Hornsey, et al., 2020; Huddy et al., 2015; Hurst & Stern, 2020). However, when the same information is attributed to an out-group member, it may be dismissed or even lead to backfire effects (Esposo et al., 2013; Dixon et al., 2017). Moreover, perceived similarity between audience and messenger was found to foster trust, liking, and willingness to act (Krantz & Monroe, 2016). Conversely, messages perceived as

* Corresponding author.
E-mail address: danielle.goldwert@nyu.edu (D. Goldwert).

https://doi.org/10.1016/j.jenvp.2025.102896
Received 28 July 2025; Received in revised form 19 December 2025; Accepted 22 December 2025
Available online 23 December 2025

identity-incongruent are likely to trigger resistance or motivated reasoning (Kahan, 2015). Thus, here we investigate how the credibility and effectiveness of climate change information varies by source type, and whether this impact is moderated by recipients' political identity. Given that in the context of climate change, the most relevant dimension of identity has been found to be political (Hornsey et al., 2016; Berkebile-Weinberg, Goldwert, et al., 2024), we specifically focus on the interactions between information sources and people's political identity in shaping climate beliefs, policy preferences, and behavioral commitments.

More recent experimental work further suggests that the effects of source credibility are not uniform but may vary depending on how the message is framed and the identity of the messenger. In particular, prior work points to the importance of congruence between a message's content and the messenger's identity in credibility judgements (Boerman et al., 2022). For example, a religious leader conveying information about environmental degradation and importance of action using a purity or sanctity of the environment frame, would be considered a congruent frame-source interaction. On the other hand, a fossil fuel company sending a message emphasizing the efficacy of climate advocacy would be considered an incongruent frame-source interaction. Such interactions were found to have either enhancing or diminishing message persuasiveness effects. For instance, military leaders or Republican elites conveying information about the importance of climate action led to enhanced persuasion among Republican audiences, whereas the same messages delivered by climate scientists elicited reduced impact (Bolsen et al., 2019). In this case the source-frame incongruence operated through the "unconventional source" effect: when in-group leaders take unexpected positions, their messages seem especially honest and credible (Benegal & Scruggs, 2018; Prior et al., 2014). In support of a congruence hypothesis however, studies suggested that climate messages attributed to a congruent source, such as a regulatory agency, increased pro-environmental intentions (Craig & McCann, 1978; Kim & Kim, 2013). Yet another line of work finds that message frames were most effective when the source was not explicitly indicated (Li & Su, 2018; Vlasceanu & Coman, 2022). These mixed findings underscore the need for systematic experimental evidence examining how information from different sources influences climate-relevant outcomes. In this paper, we use *climate information incorporation* as an umbrella term for downstream outcomes—including beliefs, policy support, policy prioritization, and behavioral commitments—that reflect the integration of climate information into belief and action tendencies (Goldwert & Vlasceanu, 2025). We discuss belief change within a belief systems perspective that represents beliefs as interdependent representations embedded in dynamic networks (Vlasceanu, Dyckovsky, & Coman, 2024). Moreover, we consider behavioral intentions/commitments as behavioral signatures of information incorporation (Goldwert & Vlasceanu, 2025).

In sum, existing research has identified that source credibility and identity alignment can shape responses to climate messages, but the direction and strength of these effects remain mixed across studies, likely due to variation in message content, source characteristics, and study designs. Despite extensive theoretical or experimental work on source effects, few studies have systematically tested how a wide variety of prevalent information sources interact with specific message frames across multiple climate-relevant outcomes. Our study addresses this gap by experimentally manipulating both source and frame in a high-powered U.S. sample to assess their independent and interactive effects on climate beliefs, policy preferences, and behavioral commitments.

The present research tests the differential impact of information sources on people's incorporation of climate information. As preregistered, a sample of 9076 U.S. residents recruited on Prolific were randomly assigned to one of 12 sources of information (i.e., scientists, weather reporters, doctors, government agencies, non-governmental organizations, corporations, grassroots advocates, military, republican political leaders, democratic political leaders, religious leaders, fossil fuel companies) or to a no-source control condition. These sources were selected according to recent work documenting the public's trust levels in climate information sources (Maibach et al., 2023). In each condition participants received information about climate change, randomly delivered in one of two theoretically grounded frames, found to be amongst the most effective frames at stimulating climate action (Goldwert, Patel, et al., 2025): (1) Collective-Efficacy-Emotional-Benefit frame, which emphasizes the efficacy of climate action as well as the personal benefits of engaging in such behaviors, and (2) Binding-Moral-Foundations, which emphasizes how preserving the purity and sanctity of the environment is a sacred moral imperative. After reading the information, participants rated the credibility of the source of information, then, in a randomized order they rated their climate beliefs, support for climate policy, climate policy prioritization, and behavioral commitments. Finally, they provided demographic information (including political identity), were debriefed, and compensated.

We preregistered three hypotheses concerning how source credibility and congruence might shape climate information incorporation (defined as changes in beliefs, policy preferences, and behavioral commitments). We hypothesized that higher perceived source credibility would be associated with stronger climate beliefs, greater policy support, increased policy prioritization, and higher behavioral commitments (H1).We also hypothesized that identity congruence (i.e., a match between the recipient's political or religious identity and the source of the message) would increase climate information incorporation, specifically by boosting climate beliefs, policy preferences, and behavioral commitments (H2). Finally, we hypothesized that source-frame congruence (i.e., alignment between a message's content and the perceived role or values of the source) would similarly increase these outcomes (H3).

1. Methods

Participants. We conducted an a priori power analysis using the software WebPower (Zhang & Yuan, 2018), and determined that in order to achieve 80 % power to detect a moderate effect size of Cohen's $D = 0.3$, at $\alpha = .002$ (resulting from adjusting for 24 multiple comparisons), in a 26-condition between-subjects design, a sample of 9100 participants would be required. A total of 9084 U.S. residents over the age of 18 completed the study on the platform Prolific. Five participants who failed an attention check ("*Please select the color purple from the list below. We would like to make sure that you are reading these questions carefully.*") and one duplicate participant ID was excluded, yielding a final analytic sample of 9076 participants ($Mage = 41.21$, $SD = 13.98$). The sample was 40.8 % male and 59.2 % female. Participants identified as Democrat (35.8 %), Republican (33.9 %), or Independent/other (30.3 %). The study was conducted following the approval of the Institutional Review Board of Stanford University.

Design and Procedure. Participants joined the Qualtrics survey posted through the Prolific platform and read the informed consent, after which they were randomly assigned to one of 26 conditions (2 frames $\times$ 13 sources). This study used a between-subjects post-test-only experimental design. No pretest measures were collected to reduce demand characteristics and ensure that exposure to the treatment was not influenced by prior measurement. In each condition participants received information about climate change, randomly delivered in one of two frames: (1) Collective-Efficacy-Emotional-Benefit frame, or (2) Binding-Moral-Foundations. The 13 sources of information randomly selected to deliver this information were: no-source, scientists, weather reporters, doctors, government agencies, non-governmental organizations, corporations, grassroots advocates, military, republican political leaders, democratic political leaders, religious leaders, fossil fuel companies. After reading the information, participants rated the credibility of the source of information, then, in a randomized order they rated their climate beliefs, support for climate policy, climate policy prioritization, and behavioral commitments. Finally, they provided demographic information (including political identity), were debriefed, and compensated.

Materials. The two frames used were adapted from Goldwert and colleagues (2025). The Collective Efficacy and Emotional Benefit intervention began by emphasizing the high prevalence and efficacy of pro-climate beliefs amongst Americans, and offered examples of effective climate actions, such as sharing information on social media, supporting climate advocacy groups, or attending climate marches. Participants were then asked to recall (or imagine) a time they felt positive emotions after participating in climate action, or a time when they formed a friendship while engaging in climate action. Finally, they were told that taking action for the climate can make them feel happier, create strong friendships, and build deep connections with others.

The Binding moral foundations frame used moral reframing to appeal to purity and emphasize the urgency of preserving America's natural wonders from pollution. The text began by highlighting America's natural beauty, calling for the protection of the purity of its landscapes (e.g., "*America is a land of pure beauty and pristine nature. We must protect these sacred wonders from pollution and degradation.*"). Participants were informed about the threats of climate change to these landscapes (e.g., "*Our pristine nature and national parks are at risk of descending into a shell of what they once were.*"). They viewed examples like the Great Smoky Mountains' polluted air, Old Faithful's potential to dry up, and the burning of Sequoia National Park's giant trees. Participants then indicated their feelings of impurity and disgust regarding these examples, reflecting on how sinful it would be to lose these natural wonders. They were then prompted to write about how they felt Americans were failing to keep these treasures pure and sacred, and were urged to take action to prevent further desecration of the nation's greatest natural wonders.

Within each frame, participants were provided the following instructions: "The following message is from *[insert source here]*." To increase external validity, we designed logos for each source, which was displayed on each page containing relevant information.

Outcome variables Source credibility was measured using the mean of four items (on a scale ranging from 0 = "*strongly disagree*" to 100 = "*strongly agree*"): "*This source is well informed/knowledgeable," "I find this source honest/reliable/trustworthy," "I believe this source cares about people like me," and "This source likely has a hidden agenda,"* (reverse-coded).

Climate information incorporation was operationalized via four preregistered outcomes: beliefs (4-item composite), policy support (9 items), policy prioritization (5 binary forced-choice pairs), and behavioral commitments (4-item composite):

Belief in climate change was measured as the mean of four items (on a scale ranging from 0 = "*not at all accurate*" to 100 = "*extremely accurate*"): *"Taking action to fight climate change is necessary to avoid a global catastrophe," "Human activities are causing climate change," "Climate change poses a serious threat to humanity," and "Climate change is a global emergency."*

Climate policy support was measured across nine continuous policy support items (on a scale ranging from 0 = "not at all" to 100 = "very much so"): *"I support raising carbon taxes on gas/fossil fuels/coal?," "I support significantly expanding infrastructure for public transportation," "I support increasing the number of charging stations for electric vehicles," "I support increasing the use of sustainable energy such as wind and solar energy," "I support increasing taxes on airline companies to offset carbon emissions," "I support protecting forested and land areas," "I support investing more in green jobs and businesses," "I support introducing laws to keep waterways and oceans clean," and "I support increasing taxes on carbon intense foods (for example meat and dairy)."*

Policy priority was measured as binary choices between five pairs of policies. Each policy pair included one climate policy and one climate-irrelevant policy *("Significantly expanding infrastructure for public transportation* (e.g. *railroad) vs. Increasing* funding *for resources and facilities in primary schools", "Increasing the number of charging stations for electric vehicles vs. Reducing caps on government discretionary spending", "Increasing the use of renewable energy such as wind and solar energy vs. Increasing the federal minimum wage", "Investing more in green jobs and businesses vs. Increasing* funding *for military equipment", and "Implementing strict carbon emission caps on airline companies vs. Expanding government subsidies for health insurance"*).

Behavioral commitments were measured using a normalized composite index of commitments to attending a climate event or demonstration, initiating a conversation about climate change, supporting a pro-climate political campaign, and sharing information online.

Political party affiliation was measured using the question: "*What political party do you identify with?*" with 4 multiple choice options (Democrat, Republican, Independent, Other).

Demographics. Participants also indicated their gender (i.e. multiple choice: *"male", "female", "prefer not to say", "Non-binary/third gender/other"*), age (in years), education level (i.e., multiple choice: *"0–6 (up to grade school)", "7–12 (up to high school)", "13–16 (college/undergraduate degree/certificate training)", "More than 17 years (doctorate degree, medical degree, etc.)", "prefer not to answer"*), political orientation for both economic and social issues (from *"extremely liberal/left-wing"* to *"extremely conservative/right-wing"*), total yearly household income (i.e., multiple choice, where income was listed in 8 steps ($5000 each), from *"Less than $10,000" to "$200,000 or more" and "Prefer not to respond"*), and subjective socioeconomic status, using the MacArthur single-item perceived rank relative to others in their group (i.e. multiple choice: *"Rung 1 (Bottom) People here are worst off"* to *"Rung 10 (Top) People here are the best off"*).

Analysis and Coding. To test identity congruence (H2), we considered political participant-source pairings as follows. Political identity was considered congruent in the case of Democratic participants receiving information from Democratic political leaders and incongruent when receiving information from Republican leaders; Republican participants receiving information from Republican political leaders was also considered congruent, Republican participants receiving information from Democratic leaders was considered incongruent. For all interaction terms (e.g., participant × source identity), variables were dummy-coded (0 = no, 1 = yes), such that interaction coefficients reflect the additive effect of matched identity.

To examine source-message congruence (H3), we considered source-frame combinations as follows. The Collective-Efficacy-Emotional-Benefit frame was coded as congruent when delivered by scientists, doctors, weather reporters, grassroots advocates, NGOs, or Democratic leaders, and incongruent in all other cases. The Binding Moral Foundations frame was coded as congruent when delivered by religious leaders, Republican leaders, or government agencies, and incongruent in all other cases.

2. Results and discussion

All effect sizes and confidence intervals are reported in Supplement.

To assess our first pre-registered hypothesis (H1; Supplement Section 1), we conducted linear regressions with source credibility as the predictor, belief, policy support, policy prioritization, and behavioral commitments, as the outcomes, statistically adjusting for political party, gender, age, education, income, and socioeconomic status variables. We found that source credibility significantly predicted belief in climate change ($\beta = 0.46$, $SE = 0.01$, $t = 55.10$, $p < .001$), policy support ($\beta = 0.43$, $SE = 0.01$, $t = 48.12$, p $< .001$), policy prioritization ($\beta = 0.31$, $SE = 0.01$, $t = 29.83$, $p < .001$), and behavioral commitment ($\beta = 0.54$, $SE = 0.01$, $t = 61.48$, $p < .001$). Participants who rated the information source as more credible were significantly more likely to endorse pro-climate beliefs, support climate policies, prioritize climate legislation, and commit to action. These findings are consistent with Vlasceanu and Coman (2022), who showed that higher source credibility was associated with greater acceptance of information and stronger belief formation in the context of COVID-19 misinformation. Together, these results suggest that source credibility exerts a robust influence across domains and may facilitate the cognitive and motivational processes involved in incorporating new information into belief and behavioral systems.

Contrary to our second hypothesis, positing a political identity congruence effect (H2; Supplement Section 2), we did not find evidence for increased message impacts in congruent (Democratic-Democratic or Republican-Republican) compared to incongruent (Democratic-Republican) pairing between participants and sources. In linear regression models predicting each outcome (belief, policy support, policy prioritization, and behavioral commitment) from participant partisanship, source partisanship, their interaction terms, and covariates (gender, age, education, ideology, income, and socioeconomic status), interaction terms between participant partisanship and matched political sources were nonsignificant across all outcomes: For belief (Democratic × Democratic: $\beta = 0.11$, $SE = 0.08$, $t = 1.25$, FDR-corrected $p = .82$; Republican × Republican: $\beta = 0.02$, $SE = 0.09$, $t = 0.23$, FDR-corrected $p = .82$), policy support (Democratic × Democratic: $\beta = 0.04$, $SE = 0.09$, $t = 0.45$, FDR-corrected p $= .82$; Republican × Republican: $\beta = -0.08$, SE $= 0.09$, $t = -0.86$, FDR-corrected $p = .52$), policy prioritization (Democratic × Democratic: $\beta = -0.02$, $SE = 0.09$, $t = -0.23$, FDR-corrected $p = .82$; Republican × Republican: $\beta = -0.10$, $SE = 0.09$, $t = -1.08$, FDR-corrected $p = .52$), and behavioral commitments (Democratic × Democratic: $\beta = -0.05$, $SE = 0.09$, $t = -0.58$, FDR-corrected $p = .82$; Republican × Republican: $\beta = -0.10$, $SE = 0.09$, $t = -1.09$, FDR-corrected $p = .52$). Similarly, cross-partisan exposure (opposite-party source) did not produce significant effects for belief ($\beta = -0.03$, $SE = 0.07$, $t = -0.42$, FDR-corrected $p = .90$), policy support ($\beta = -0.08$, $SE = 0.07$, $t = -1.16$, FDR-corrected $p = .84$), policy prioritization ($\beta = 0.00$, $SE = 0.07$, $t = 0.02$, FDR-corrected $p = .99$), or behavioral commitments ($\beta = -0.06$, $SE = 0.07$, $t = -0.81$, FDR-corrected $p = .84$). These results diverge from a well-documented literature showing that identity-congruent sources, particularly those aligned with recipients' political ideology, can increase message persuasiveness on attitudinal measures (e.g., Fielding et al., 2020). Our findings do not replicate this attitudinal effect, nor do they support downstream effects on behavioral intentions or policy support. However, these results align with prior research that incorporated behavioral measures, for which the source identity hypothesis was also not supported (Hurst & Stern, 2020; Vlasceanu & Coman, 2022). Together, these findings suggest that the persuasive effects of identity-congruent sources may be attenuated in the context of highly polarized issues like climate change, particularly when participants are already familiar with the content, or when exposure occurs outside of high-stakes or interpersonal settings. This points to boundary conditions for source-recipient alignment effects, which may depend on message novelty, perceived personal relevance, or contextual factors such as information overload and issue saturation.

Also contrary to the source-frame congruence prediction (H3; Supplement Section 3), we found no effect of congruence on belief ($p = .288$), policy support ($p = .210$), policy priority ($p = .704$), or behavioral commitments ($p = .442$). This result did not provide support for the 'unconventional source' effect (Benegal & Scruggs, 2018), positing that unexpected or counter-stereotypical messengers can enhance persuasiveness. However, given that this effect had been documented in other contexts, it is possible that the null results found here reflect boundary conditions specific to climate messaging (e.g., entrenched beliefs, perceived source motivation, or general skepticism).

In additional pre-registered exploratory analyses, we examined how credible climate information was perceived when communicated by different sources (Supplement Section 4). We conducted a linear regression with credibility as the dependent variable and information source as the independent variable, controlling for party identification, gender, age, education, ideology, income, and socioeconomic status. Bonferroni-adjusted pairwise comparisons revealed that climate information was perceived as significantly more credible when communicated by scientists ($M = 69.7$, $SE = 0.93$, $p < .001$) and grassroots advocates ($M = 69.5$, $SE = 0.94$, $p = .0012$), relative to a no-source control ($M = 64.3$). The least credible sources were Republican political leaders ($M = 59.6$, $SE = 0.94$, $p = .0059$) and fossil fuel companies ($M = 60.0$, $SE = 0.96$, $p = .0202$) (Fig. 1A). These findings suggest that scientists and grassroots community advocates are perceived as some of the most credible climate information sources. While our study did not find that source alone directly influenced belief or behavior change, prior work has shown that messages from trusted scientific or community-based sources can enhance message acceptance and persuasion, particularly when combined with aligned values or efficacy-based content (e.g., Bolsen et al., 2019; Brewer & Ley, 2013; Fielding et al., 2020).

When including party identification (Democrat, Republican, Independent) as moderator in the model, we found a significant source by ideology interaction ($F(24, 8981) = 2.57$, $p < .001$). Bonferroni-adjusted pairwise comparisons revealed that for Democrats, messages attributed to Republican sources ($M = 55.81$, $p < .001$) and fossil fuel companies ($M = 60.48$, $p = .0049$) were rated as significantly less credible than the no-source control ($M = 68.38$) (Fig. 1B). Among Republicans, no source differed significantly from control after correction (Fig. 1B). These findings suggest that across the political spectrum there do not appear to be stark differences in climate information source credibility, beyond Democrats being skeptical of Republican leaders and fossil companies, which is not surprising given the well documented denialism and solution opposition by these actors (Oreskes & Conway, 2011; Stokes, 2020).

Also in preregistered exploratory analyses, we tested whether different information sources differentially influenced beliefs, policy support, policy priority, or behavioral outcomes (Supplement Section 5). We fit linear regressions with source as the predictor, controlling for message frame. As message framing (Collective Efficacy + Emotional Benefit vs. Binding Moral Foundations) did not significantly affect climate belief ($p = .506$), policy support ($p = .656$), or policy priority ($p = .295$), we collapsed across frames for these outcomes. The overall effect of source was non-significant for beliefs ($p = .336$), policy support ($p = .365$), or policy prioritization ($p = .349$). Bayesian ANOVAs provided extreme evidence against an effect of source on any outcome. Bayes factors were far below 1, indicating that the intercept-only (null) model was overwhelmingly favored for belief in climate change (BF $= 1.34 \times 10^{-7}$), policy support (BF $= 1.38 \times 10^{-7}$), policy prioritization (BF $= 1.23 \times 10^{-7}$), and behavioral commitment (BF $= 1.17 \times 10^{-7}$). While frame ($p < .001$) did significantly impacted behavioral commitments, suggesting that only the Collective-Efficacy-Emotional-Benefit frame increased commitments, when conducting follow-up analyses within this condition, we found that information sources significantly influenced commitment ($p = .024$), but no pairwise differences remained significant after Bonferroni correction for multiple comparisons. These findings remained consistent when including participants' political ideology as a moderator (Fig. 2). Further, to ensure that effects were not influenced by inattentive responding, we repeated all primary pre-registered analyses including only participants who passed the checks assessing comprehension of the message content. Results were unchanged (Supplementary Information, Section 6). Thus, we do not find evidence that information sources lead to the differential incorporation of climate change messages into belief systems, policy preferences, or behavioral commitments, suggesting that efforts to influence these outcomes should focus on alternative messaging features.

Conclusion and Future Directions. Together, these findings highlight an important distinction between credibility judgments and information incorporation. Across conditions, participants were sensitive to source cues and adjusted their evaluations of source credibility accordingly, indicating that source information was noticed and meaningfully processed. However, these credibility judgments did not translate into changes in climate beliefs, policy support, or behavioral commitments. This dissociation is consistent with dual-process and information-integration frameworks, which posit that evaluative

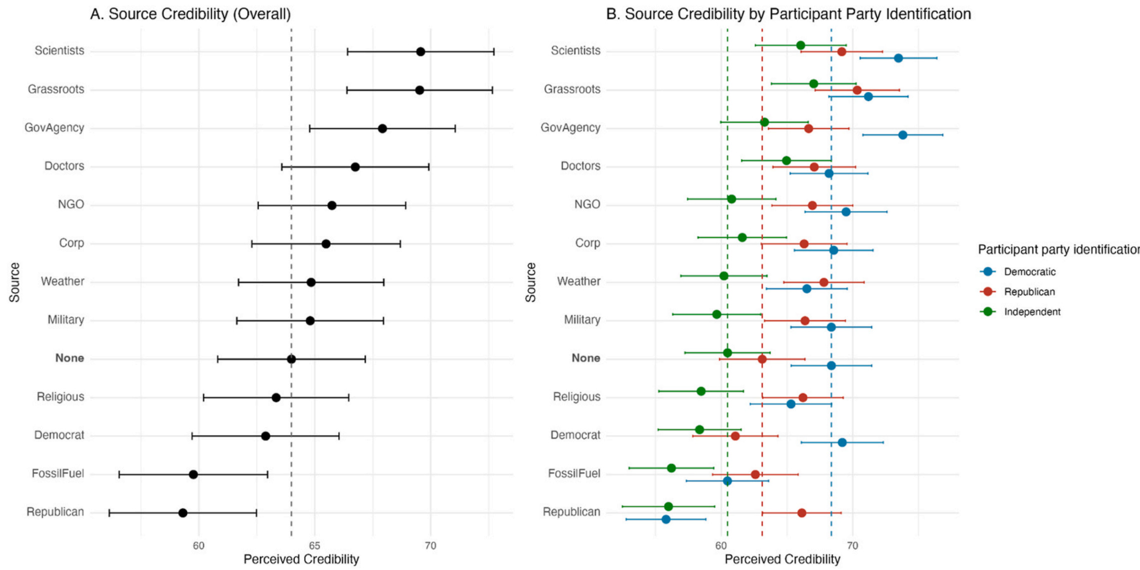

Fig. 1. Perceived credibility of 12 information sources compared to a no-source control condition of a sample of 9076 U.S. residents (Panel A); Perceived credibility of 12 information sources compared to a no-source control condition of the same sample as Panel A, but split by self-reported political party identification: Democratic (Blue), Republican (Red), or Independent (Green) (Panel B). Error bars represent 95 % CI. Vertical dotted lines indicate the means in the control condition within each group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

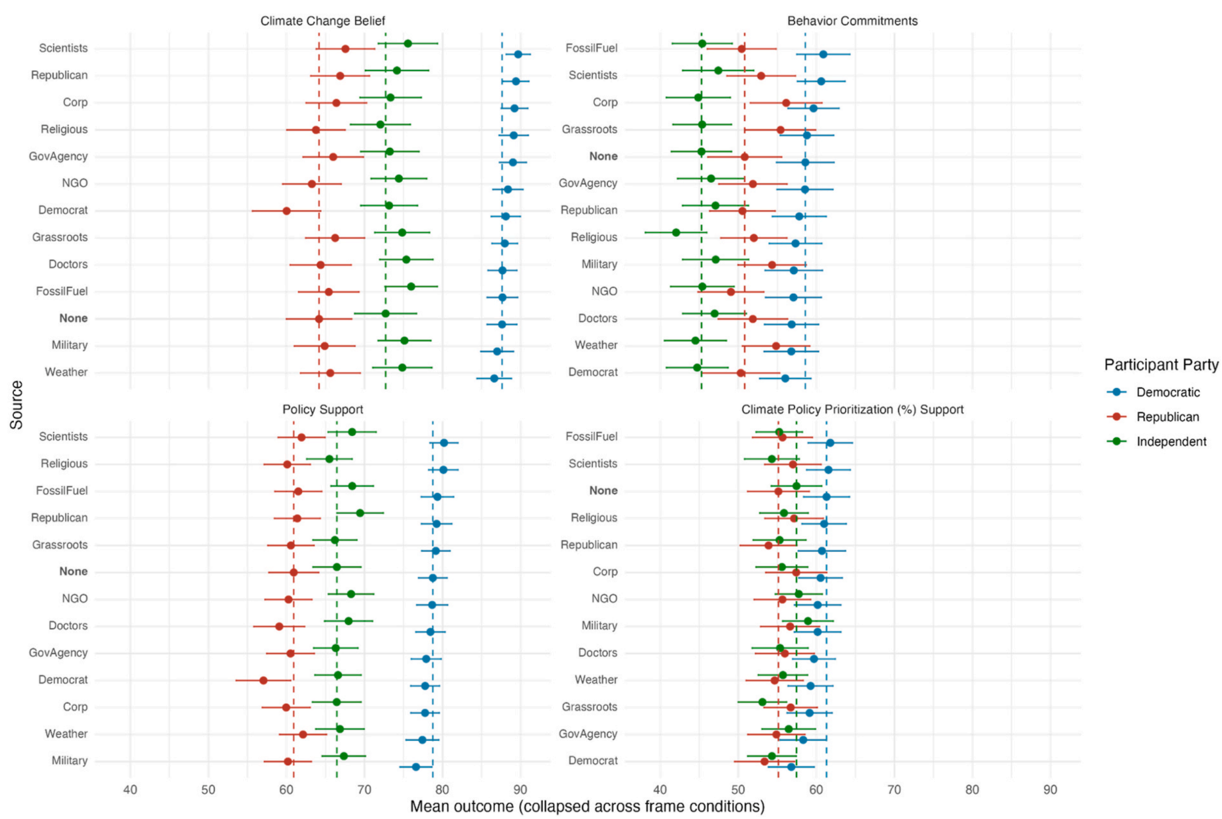

Fig. 2. The effects of information source on the beliefs (A), commitments (B), policy support ©, and policy prioritization (D) of Democratic (blue), Republican (Red), or Independent (Green) participants. Error bars represent 95 % CI. Vertical dotted lines indicate the means in the control condition within each political group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

judgments about a message or its source can occur without necessarily triggering deeper belief revision or attitude change, particularly when recipients hold relatively crystallized or identity-linked views (Petty & Cacioppo, 1986).

Several study-specific and contextual factors may help explain the absence of downstream source effects. Baseline climate beliefs in the present U.S. sample were relatively high, potentially limiting sensitivity to source-based persuasion through ceiling effects. In addition, the politicized nature of climate change may constrain the influence of source cues via motivated reasoning. Source effects may also depend on message novelty or diagnosticity; credibility cues may matter most when information is sufficiently surprising to prompt heightened source scrutiny. Further, the present design employed a single exposure to the source-accompanied interventional content; however, individuals are typically exposed to such information repeatedly, over a longer time-course, allowing time for socio-cognitive processes to solidify into observable patterns of behavior (Vlasceanu et al., 2023). Finally, while our outcomes captured belief endorsement, policy support, and behavioral intentions, more fine-grained measures of belief uncertainty or ambivalence may be more sensitive to subtle source effects.

Importantly, these null effects should not be interpreted as evidence that information sources do not matter, but rather as evidence that the persuasive influence of source credibility is conditional on features of the message, the recipient, and the broader informational context. Future research should examine how source effects unfold under conditions of repeated exposure, greater message novelty, or social transmission, including whether different sources shape how climate information is shared and propagated through social networks. Clarifying these conditions will be essential for understanding when and how information sources contribute to meaningful climate engagement.

CRediT authorship contribution statement

Danielle Goldwert: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Madalina Vlasceanu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Funding

This work was supported by National Science Foundation (NSF) Graduate Research Fellowship (GRF) Grant No. DGE-2039655 (to D.G.)

Declaration of competing interest

None.

; .

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2025.102896.

Data availability

All data and code are publicly available through the Open Science Framework (osf.io/hta54/overview).

References

Benegal, S. D., & Scruggs, L. A. (2018). Correcting misinformation about climate change: The impact of partisanship in an experimental setting. *Climatic Change, 148*, 61–80. https://doi.org/10.1007/s10584-018-2192-4

Berkebile-Weinberg, M., Goldwert, D., Doell, K. C., Van Bavel, J. J., & Vlasceanu, M. (2024). The differential impact of climate interventions along the political divide in 60 countries. *Nature Communications, 15*(1), 3885.

Boerman, S. C., Meijers, M. H., & Zwart, W. (2022). The importance of influencer-message congruence when employing greenfluencers to promote pro-environmental behavior. *Environmental Communication, 16*(7), 920–941.

Bolsen, T., Palm, R., & Kingsland, J. T. (2019). The impact of message source on the effectiveness of communications about climate change. *Science Communication, 41*(4), 464–487. https://doi.org/10.1177/1075547019863154

Brewer, P. R., & Ley, B. L. (2013). Whose science do you believe? Explaining trust in sources of scientific information about the environment. *Science Communication, 35*(1), 115–137.

Chaiken, S. (1980). Heuristic versus systematic information processing and the use of source versus message cues in persuasion. *Journal of personality and social psychology, 39*(5), 752.

Cohen, G. L. (2003). Party over policy: The dominating impact of group influence on political beliefs. *Journal of personality and social psychology, 85*(5), 808.

Craig, C. S., & McCann, J. M. (1978). Assessing communication effects on energy conservation. *Journal of Consumer Research, 5*(2), 82–88.

Dixon, G., Hmielowski, J., & Ma, Y. (2017). Improving climate change acceptance among U.S. conservatives through value-based message targeting. *Science Communication, 39*(4), 520–534. https://doi.org/10.1177/1075547017715473

Esposo, S. R., Hornsey, M. J., & Spoor, J. R. (2013). Shooting the messenger: Outsiders critical of your group are rejected regardless of argument quality. *British Journal of Social Psychology, 52*, 386–395.

Fielding, K. S., & Hornsey, M. J. (2016). A social identity analysis of climate change and environmental attitudes and behaviors: Insights and opportunities. *Frontiers in Psychology, 7*, 121.

Fielding, K. S., Hornsey, M. J., Thai, H. A., & Toh, L. L. (2020). Using ingroup messengers and ingroup values to promote climate change policy. *Climatic Change, 158*(2), 181–199.

Goldwert, D., Constantino, S. M., Patel, Y., Sabherwal, A., Semken, C., Brick, C., Castiglione, A., Debnath, R., Doell, K. C., Dubey, R., Fang, K., Goldberg, M. H., Ma, W. J., Nielsen, K. S., Rathje, S., Schneider, C. R., Sheldrick, M., Shreedhar, G., van der Linden, S., … Vlasceanu, M. (in press). A megastudy of behavioral interventions to catalyze public, political, and financial climate advocacy. PNAS Nexus.

Goldwert, D., Patel, Y., Nielsen, K. S., Goldberg, M. H., & Vlasceanu, M. (2025). Climate action literacy interventions increase commitments to more effective mitigation behaviors. *PNAS Nexus, 4*(6). pgaf191.

Goldwert, D., & Vlasceanu, M. (2025). An individual–collective–systemic behavioral climate intervention framework. *Current Opinion in Behavioral Sciences, 65*, Article 101569.

Hart, P. S., & Nisbet, E. C. (2012). Boomerang effects in science communication: How motivated reasoning and identity cues amplify opinion polarization about climate mitigation policies. *Communication Research, 39*(6), 701–723. https://doi.org/10.1177/0093650211416646

Hornsey, M. J., Harris, E. A., Bain, P. G., & Fielding, K. S. (2016). Meta-analyses of the determinants and outcomes of belief in climate change. *Nature Climate Change, 6*(6), 622–626.

Hovland, C., & Weiss, W. (1951). The influence of source credibility on communication effectiveness. *Public Opinion Quarterly, 15*, 635–650. https://doi.org/10.1086/266350

Huddy, L., Mason, L., & Aarøe, L. (2015). Expressive partisanship: Campaign involvement, political emotion, and partisan identity. *American Political Science Review, 109*(1), 1–17. https://doi.org/10.1017/S0003055414000604

Hurst, K., & Stern, M. J. (2020). Messaging for environmental action: The role of moral framing and message source. *Journal of Environmental Psychology, 68*, Article 101394.

Kahan, D. M. (2015). The politically motivated reasoning paradigm. In R. A. Scott, & S. M. Kosslyn (Eds.), *Emerging trends in the social and behavioral sciences*. New York, NY: Wiley Online. https://doi.org/10.1002/9781118900772.etrds0417.

Kahan, D. M., Jenkins-Smith, H., & Braman, D. (2011). Cultural cognition of scientific consensus. *Journal of Risk Research, 14*(2), 147–174.

Kim, S.-B., & Kim, D.-Y. (2013). The effects of message framing and source credibility on green messages in hotels. *Cornell Hospitality Quarterly, 55*(1), 64–75. https://doi.org/10.1177/1938965513503400

Krantz, S. A., & Monroe, M. C. (2016). Message framing matters: Communicating climate change with forest landowners. *Journal of Forestry, 114*(2), 108–115.

Li, N., Su, & Yi-Fan, L. (2018). Message framing and climate change communication: A meta-analytical review. *Journal of Applied Communications, 102*(3). https://doi.org/10.4148/1051-0834.2189

Maibach, E. W., Uppalapati, S. S., Orr, M., & Thaker, J. (2023). Harnessing the power of communication and behavior science to enhance society's response to climate change. *Annual Review of Earth and Planetary Sciences, 51*(1), 53–77.

Oreskes, N., & Conway, E. M. (2011). *Merchants of doubt: How a handful of scientists obscured the truth on issues from tobacco smoke to global warming*. Bloomsbury Publishing USA.

Petty, R. E., & Cacioppo, J. T. (1986). *Communication and persuasion: Central and peripheral routes to attitude change* (1st ed.). Springer. https://doi.org/10.1007/978-1-4612-4964-1

Petty, R. E., Kasmer, J. A., Haugtvedt, C. P., & Cacioppo, J. T. (1987). Source and message factors in persuasion: A reply to stiff's critique of the elaboration likelihood model. *Communication Monographs, 54*(3), 233–249. https://doi.org/10.1080/03637758709390229

Prior, J., Partridge, E., & Plant, R. (2014). 'We get the most information from the sources we trust least': residents' perceptions of risk communication on industrial contamination. *Australasian Journal of Environmental Management, 21*(4), 346–358.

Stokes, L. C. (2020). *Short circuiting policy: Interest groups and the battle over clean energy and climate policy in the American States*. Oxford University Press.

Vlasceanu, M., & Coman, A. (2022). Information sources differentially trigger Coronavirus-related belief change. *International Journal of Data Science and Analytics, 13*, 287–298.

Vlasceanu, M., Doell, K. C., Bak-Coleman, J. B., Todorova, B., Berkebile-Weinberg, M. M., Grayson, S. J., Patel, Y., Goldwert, D., Pei, Y., Chakroff, A., et al. (2024). Addressing

climate change with behavioral science: A global intervention tournament in 63 countries. *Science Advances, 10*(6). eadj5778.
Vlasceanu, M., Dyckovsky, A. M., & Coman, A. (2024). A network approach to investigate the dynamics of individual and collective beliefs: Advances and applications of the bending model. *Perspectives on Psychological Science, 19*(2), 444–453.
Zhang, Z., & Yuan, K. H. (2018). In *Practical Statistical Power Analysis Using Webpower and R*. Granger, IN: ISDSA Press. https://webpower.psychstat.org.